Lai Wei

iaml.ai linkedin.com/in/lai-wei-roy github.com/roywei

Email: royweilai@gmail.com
Mobile: 412-953-9889

Education

- **Carnegie Mellon University** *Pittsburgh, PA*
  Master of Science: Electrical and Computer Engineering *2015-Jan – 2016-May*
- **Nanyang Technological University** *Singapore*
  Bachelor of Engineering: Electrical and Electronic Engineering *2010-Aug – 2014-May*

Work History

- **Meta Superintelligence Lab** *2024-Dec – Present*
  Engineering Manager, Inference Engine
  - Scaled inference engine team from 9 to 16 engineers; led model enablement and inference optimizations. Mentored 2 team members into **vLLM committers** within 6 months.
  - Led inference for the **Meta Muse Spark** launch, delivering offline inference, RL rollout, and production serving optimizations cross-functionally with research and product partners.
  - Drove inference optimizations spanning speculative decoding, CPU KV-cache offload, **batch-invariant inference** (bit-level parity with the RL trainer and sampler), MoE load balancing, and parallelism strategies. Pushed SOTA Blackwell perf on vLLM (DSR1 on GB200) and shipped Llama 4 day-zero on vLLM.
  - Drove company-wide adoption of **vLLM** as the unified inference engine for open-source models across NVIDIA, AMD, TPU, and MTIA on multiple cloud capacities — >**80% of inference workload migrated**; advanced the move toward Docker, OSS, and cloud-native infrastructure.
  - Built internal inference agents (“Claw”) that automate debugging, deployment, and issue triage for internal customers; built capacity-estimation and perf-sweep tooling for model launches.
  - Co-led ecosystem partnerships with NVIDIA, AMD, AWS, PyTorch, and the vLLM community; secured AWS partnership credits for the **vLLM router** launch.
- **Amazon Web Services (AWS) SageMaker** *2018-Jan – 2024-Nov*
- *Software Development Manager* *2023-Dec – 2024-Nov*
  - Led a team of 7 to optimize PyTorch for next-generation compute instances, including H200, L40S GPUs, AWS Graviton CPUs, and upcoming GH/B200 instances.
  - Collaborated cross-company with Meta, Nvidia, EC2, Annapurna Labs, and SageMaker on PyTorch performance at large-cluster scale; launched Meta’s SAM v2 model on SageMaker infra.
  - Directed development of AWS Deep Learning Containers, integrating SageMaker DataParallel/Model Parallel libraries; pioneered AI-agent security patching automation, saving an estimated 50 SDE-weeks per year.
- *Senior Software Development Engineer / Tech Lead Manager* *2022-Oct – 2023-Nov*
  - Established a specialized team to ship **AWS-optimized PyTorch** with out-of-the-box support for AWS EFA and enhanced NCCL; raised network utilization from 70% to 90% on A100 clusters — adopted by AWS’s largest LLM training customers.
  - Redesigned the SageMaker Distributed Data Parallel Library to integrate with PyTorch as a custom distributed backend; achieved a **40% training-time reduction** for GPT NeoX, DALL-E, and guided diffusion models.
- *Software Development Engineer* *2018-Jan – 2022-Sep*
  - Spearheaded loss-function optimization for the **AWS MLPerf 2020** benchmark, enabling convergence at extremely large batch sizes — trained MaskRCNN on 512 GPUs in <7 minutes and T5-3B on 2048 GPUs in 4.68 days.
  - Built **Deep Java Library (DJL)** for framework-agnostic Java inference (TensorFlow / TF-Lite engines); Apache MXNet committer and primary maintainer of Keras-MXNet.
- **Cheetah Mobile America AI Lab** *2016-Jun – 2017-Oct*
  Software Engineer
  - Designed the User2Vec feature pipeline that improved Ad Click-Through Rate (CTR) prediction accuracy.
  - Constructed data pipelines generating 5M user profiles powering personalized news recommendation.

Publications & Patents

- **US Patent 11763154** — Machine Learning Service with Pre-Trained Models
- **US Patent 11769035** — Automatically Determining Configurations for Executing Recurrent Neural Networks
- **Transfer Learning for Personalized Content and Ad Recommendation**, Industry Talk on RecSys 2017
- **Machine learning approach for shaft crack detection through acoustical emission signals**, 2015 IEEE 20th Conference on Emerging Technologies & Factory Automation (ETFA)